CH 3 - Random Variables

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3.1 Random Variables

> 📘 **Definition — Random variables**
>
> A **random variable** is a fucntion X that maps points in a smaple space to real numbers
>
> $$X : S \to \mathbb{R}$$
>
> A random variable can be discrete or continuous depending on whether the **range** is countable or uncountable

Example

Roll 3 fair 6-sided dice, $S = \{(1,1,1), ..., (6,6,6)\}$, 216 pairs

List a few possible random variables

$$X = \text{sum be the 3 dice, range} = \{3, ..., 18\},$$
$$Y = \text{max of 3 dice, range} = \{1, 2, 3, 4, 6\}$$
$$Z = \text{product of the 3 dice, range} = \{1, ..., 216\}$$

Continuous random variables could include anything measured in time, temperature, heightes, weights, etc

> 📘 **Definition — Probability Mass Function**
>
> The **PMF** of a discrete random variable $X$is
>
> $$\begin{aligned} f(x) &= P(X = x), \quad \forall x \in \text{range } X \\ &= \sum P(a_i) \quad \text{sum all the probability of the sample points that map to x} \end{aligned}$$

Examples

1. $X = \text{sum of 3 dice}$

   $$f(x) = \begin{cases} \frac{1}{216} \text{ if } x{=}3 \\ \frac{3}{216} \text{ if } x{=}4 \\ ... \\ \frac{1}{216} \text{ if } x{=}18 \end{cases}$$

2. $W = 3 \text{ diigt number}$

   $$f(w) = \frac{1}{216}$$

3. $Y = \text{max of 3 dice}$

   $$f(y) = \begin{cases} \frac{1}{216} \text{ if } y{=}1 \\ \frac{7}{216} \text{ if } y{=}2 \\ ... \end{cases}$$

Note: for any pmf, $f(x) > 0$ and $\sum\left(f_{x_i}\right) = 1$ because the random variable has to take on of the values

📘 **Definition — Probability Mass Function**

📘 **Definition — Cumulative Density Funciton**

3.2 Functions of Random Variables

In practice, we may be interested in some function of a random variable rather than the random variable itself.

i.e. total revenue based on units sold

If $Y = g(X)$, we may want the pmf of Y.

Example:

$X =$ number of 6 on 3 dice

Let $Y = \begin{cases} 0 \text{ if } x=0 \\ 1 \text{ if } x=1 \text{ or } 2 \\ 2 \text{ if } x=3 \end{cases}$, with range$\{0, 1, 2\}$

Y	0	1	2
$f_Y(y)$	$\frac{125}{216}$	$\frac{90}{216}$	$\frac{1}{216}$

$f_Y(0) = P(Y = 0) = P(X = 0) = f_X(0)$

$f_Y(1) = P(Y = 1) = P(X = 1 \text{ or } X = 2) = f_X(1) + f_X(2)$

$f_Y(2) = f_X(3)$

💡 **Tip** — Interchanging probability mass function variables

In general, if $Y = g(X)$ for a discrete random variable X

$$f_Y(y) = \sum_{\text{all } x \in \text{ range s.t. } g(x)=y} f_X(x)$$

We can view that as

$$f_Y(y) = P(Y = y) = P(X \text{ tales a value that } g(x) \to y) = \sum_{g(x)=y} P(X = x)$$

Example:

At a BINGO event, players pay 14 plus 2 per “Tonnie pot” bets(maximum bets be 2).

Let $X =$ the number of Toonie bets a player

X	0	1	2
$f_X(x)$	0.4	0.1	0.5

Let $Y =$ amount paid by random player, then $Y = 14 + 2X$

Y	14	16	18
$f_Y(y)$	0.4	0.1	0.5

Definition — Proposition

If $Y = g(X)$ is a **one-to-one** function on X with inverse $g^{-1}(Y) = X$ then

$$f_Y(y) = f_X(g^{-1}(y))$$

3.3 Expectation of Random Variables

Definition — Mean, Median, Mode

Mean is average

Median is the value where half are above and half are below

Mode is the most reoccurring data

1. Let $X =$ number of kids in a family,

   we have the sample data:

X	1	2	3	4	5+
frequency	28	81	26	5	10
proportion %	18.67	54	17.33	3.33	6.67

   Mean ≈ 2.25

   Median = Mode = 2

2. Let $X =$ number of courses

   we have the sample data:

X	1 or 2	3	4	5	6+
proportion %	1	1	12	72	13

   Median = Mode = 5

   Mean ≈ 5

We can rewrite the first example mean as $\sum_{i=1}^{5} x \cdot \text{proportion with value } x$

Definition — Expectation of Random Variable

The **Expected Value** of a discrete random variable

$$E[x] = \mu = \sum_{\text{all } x \in \text{ range}} x f(x)$$

Example:

Refer to the previous example.

According to StatCan, the pmf of $X = \text{number of kids}$ is

X	1	2	3	4
$f(x)$	0.44	0.36	0.17	0.03

$E[x] = 1.79$